

Goal

Make the last 8 outputs of the perceptron equal:

01110000

This is the binary representation of the ASCII character:

'p'

Perceptron Rule

$wx + b \geq 0 \rightarrow 1$ (fires) $wx + b < 0 \rightarrow 0$ (stays quiet)

Constraints:

- Input range: [-10, 10]
 - Cannot repeat the same number consecutively.
 - Last 8 outputs are tracked.
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Step 1: Find Inputs Producing 0 and 1

Input Output ----- -5 0 1 0 2 1 3 1

Therefore:

0-producing values: -5, 1

1-producing values: 2, 3

Step 2: Desired Bit Pattern

Target:

01110000

Corresponding outputs:

0 1 1 1 0 0 0 0

Step 3: Input Sequence Used

Input Output ----- -5 0 2 1 3 1 4 1 0 0 -1 0 -2 0 -3 0

Resulting outputs:

01110000

Terminal Output

[16/128] x> -3 Perceptron stays quiet. Recent outputs (8/8): 01110000

Pattern matched! ASCII 'p' unlocked. Here is your flag:

academy{n3ur0n_m3t_227803d1}

Flag

academy{n3ur0n_m3t_227803d1}

Key Idea

1. Find one value producing 0.
2. Find one value producing 1.
3. Construct the desired binary pattern.
4. Alternate inputs producing the same bit to avoid consecutive repeats.
5. The last 8 outputs are interpreted as an ASCII character.

This challenge demonstrates how a simple 1D perceptron acts as a binary threshold classifier.